MIL-C-18021B(SHIPS)
2 May 1962
SUPERSEDING
MIL-C-18021A(SHIPS)
1 April 1958

MILITARY SPECIFICATION

CONVERTER, SPEED, UNDERWATER LOG

1. SCOPE

1.1 This specification covers speed coverters for use on Naval Ships having equipment installed which requires a speed input different from that transmitted by the underwater log system. The speed converters receives an input from the underwater log system and coverts it to the type of signal required for the equipment receiving the signal.

1.2 Classification - Speed converters shall be classed according to input and output speeds and the types of synchros used as shown in Table 1.

Table I

Input	0-40 Knots, 60 cycle 18CT6b per 360 degrees rotation							0-40 Knots, 400 cycle, 18 CT 4a	
Output Data	60 cycle			400 cycle					
Type synchro	31TX6	31TX6	31TX6	23CX4	23CX4	23TX4	23CX4	23CX4	23CX4
Range Knots per 360 degrees rotation	0-10	0-25	0-100	0-10	0-100	0-40	0-40	0-10	0-100
Model									
YC1	X		X						
YC2				X	X				
YC3	X		X	X	X				
YC4	X		X	X	X	X			
YC5				X	X	X			
YC6	X		X			X			
YC7			X	X	X	X			
YC8		X		X	X	X			
YC9		X	X	X	X	X			
YC10								X	X
YC11		X	X	X	X				
YC12	X		X	X	X		X		
YC13*	X		X	X	X	X	*(25 k/rev. on 18CT6 input)		

2 APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment For Naval Shipboard Use.

MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements for

MIL-M-15071 - Manuals, Equipment and Systems

MIL-E-17555 - Electronic and Electrical Equipment and Assoicated Repair Parts, Preparation for Delivery of.

MIL-S-20708 - Synchros, 60 and 400 Cycle, General Specification.

FSC 6605

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:54Z
Check the source to verify that this is the current version before use.

MILITARY (Cont'd.)
MIL-E-22843 - Equipment, Low Noise Level, (Naval Shipboard Use).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction unit. - When ordered under a contract schedule, a preproduction unit of the underwater log speed converter shall be furnished and fully tested by the manufacturer for performance of the tests specified herein (see 4.2 and 6.1).

3.2 Description - The speed converter shall be designed to receive a single speed synchro speed signal on a synchro control transformer, amplify the signal, and transmit synchro speed signals as required on control transmitters or torque transmitters at a definite ratio to the received signal in accordance with the model ordered. The speed converter shall indicate for system alignment purposes the received signal. The converter shall utilize amplifiers of either (1) the all magnetic type, or (2) the transistorized type, and shall be designed for continuous operation. Electron tubes shall not be used.

3.3 General features - The equipment shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein. (Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirements of this specification shall govern.)

General information
Definitions
Materials, general
Stressed or vital parts
Substitution of (equal or superior) materials
Fungus-inert materials
Unacceptable materials
Acceptable materials
Flammable materials
Arc-resistant materials
Toxic Materials
Wood
Aluminum
Magnesium
Iron or steel
Brass
Bronze
Copper
Copper-beryllium alloy
Copper-nickel alloy
Nickel-copper alloy (Monel)
Copper-nickel-zinc alloy (nickel silver)
Zinc
Other metals
Springs
Plastics
Ceramics
Glass
Lubricants and lubrication
Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Gaskets
Dials and pointers
Dial sizes
Shock, vibration and inclination (vital vibration test required)
Primary power supply circuits
Power supply tolerances (see 1.2)
Personnel protection
Shielding and radio frequency noise reduction
Ground potential and grounding
Soldering
Electrical parts mounting
Internal subassembly connection (Test cable required)
Locking devices
Washers
Ball bearings
Parts-electrical-general
Electron tube or capacitor sockets
Capacitors
Variable resistors
Transformers
Relays
Synchros
Electrical tapes
Switches
Fuses
Fuseholders and fuseclips
Metallic rectifiers
Enclosures-general
Enclosure-accessibility
Enclosure-degree of (see 3.5.1)
Enclosure-mounting (see 3.5.1)
Stiffening grooves
Minimum sheet metal thicknesses
Through bolting
Stuffing tubes
Ventilation
Size (for submarines)
Threaded devices
Rounded corners and edges
Internal subassembly protection
Drilled and tapped holes
Welding
Temperature and humidity (Special cold limit applies)
Accelerated life

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:54Z
Check the source to verify that this is the current version before use.

Acceptance inspection
Detailed test procedures
Terminal boards connectors and terminals
Synchro connections and markings
Wiring
Color coding
Electrical insulation
Dielectric strength and insulation resistance
Drawings-general
Drawings-preliminary
Drawings-working
Schematic diagrams
Wiring diagram
Drawing list
Assembly drawings
List of onboard repair parts
Drawings-manufacturing
Bill of materials
Interchangeability and standardization
Technical manuals
Repair parts
Repair parts (electronic)
Designation and marking (Serial numbers and required)
Design for production
Reports
Nomeclature
Workmanship and general examination
Operating test
General examination
Supply line voltage and frequency variation

3.4 Detail requirements -

3.4.1 Enclosure - The speed converter shall be housed in a dripproof case which shall not exceed 11 inches wide by 14 inches high by 9 inches deep, and shall be designed for bulkhead mounting. The gross weight shall not exceed 45 pounds.

3.4.2 Operation. - The speed converter shall receive synchro speed signals on the control transformer specified in table I. This input shall be converted to speed signals having the required output indicated in Table I according to the model(s) ordered, through a suitable motor driven gear train. The stability of the servo system, M_o, or maximum ratio of the output signal to the input signal, shall not exceed 1.3 and shall occur at a frequency of not less than 3 cycles per second. For this test the input signal shall be sinusoidal and the maximum rate of change need not exceed 0.55 knots per second. The saturation speed shall be greater than 1.25 knots per second. The servo shall follow the input signal in direction of travel to avoid false nulls in the 0-100K output signal.

3.4.3 Power supply. - The speed converter shall be designed to operate within specified accuracy limits on ships normal power supply of 115 volts, 60 cycles, plus or minus 7 volts and plus or minus 5 cycles per second. Excitation of the 400 cycle transmitting synchros will be 115 volts, 400 cycles, plus or minus 7 volts and plus or minus 25 cycles per second.

3.4.4 Accuracy. - The total error in the system due to static friction plus back lash between input and output shall not exceed 0.05 knot.

3.4.5 Dynamic response - The dynamic accuracy or velocity figure of merit, k_v, shall be equal to or better than 0.1 knot when speed is changing at the rate of 1.0 knots per second.

3.4.6 Air-borne and structure-borne noise. - Air-borne and structure-borne noise limits shall not exceed that specified in MIL-E-22843 for Grade A equipment.

3.4.7 Onboard repair parts. - The following items, if incorporated in the equipment, shall be furnished as onboard repair parts

Name	Quantity
Amplifier (Magnetic or transistorized)	One
Cable Connectors	Two of each type and size
Precision and special resistors	One of each type and size
Test cable	One of each type required to service equipment

3.4.8 Electronic assemblies and subassemblies such as the servo amplifier shall not be potted, embedded, or encapsulated, and shall be repairable. All components shall be capable of operating at 150°F for 125 hours continuously without failure or malfunctions.

3.4.9 Transistors, diodes, and other semiconductor devices shall be protected against electrical voltage surges of 100 percent over voltage persisting for 1 millesecond.

3.4.10 The equipment weight and center of gravity shall be included on the outline and mounting dimension plans.

3.4.11 Detail manufacturing plans shall include all dimensions and tolerances and shall completely identify all items and sources of supply.

3.4.12 Unless otherwise specified, all plans shall be "Class A" in accordance with paragraph 3.1 of MIL-D-963.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:54Z
Check the source to verify that this is the current version before use.

3.4.13 All synchros shall be contractor furnished material in accordance with MIL-S-20708.

3.4.14 Technical manuals shall be class A in accordance with MIL-M-15071.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 When specified in the order, preproduction sample of the underwater log speed converter equipment conforming in every respect to the specifications herein, and manufactured with tools, materials and methods which are the same as, or representative of, those which will be used in the manufacture of the production equipment to be furnished, shall be fully tested (all groups A B and C tests per 4.4) by the contractor at his own expense and under the supervision of the inspector to determine compliance with the aforesaid specifications The contractor shall notify in writing the cognizant inspector of the place where, and the time when, such tests will be conducted at least two weeks prior to the commencement thereof and shall attach a schedule of the tests to be conducted with such notice In addition to any other requirements of the Government with respect to the performance of the tests, the contractor shall make a record of all data obtained during such test in such form as is consonant with good engineering practice and shall furnish to the cognizant inspector three copies of all such data.

4.3 General - The tests to be performed on the equipment shall be as specified in 4.4. The attention of the manufacturer is invited to the applicable paragraphs of MIL-I-983 for information concerning these tests such as where conducted failures, test conditions and methods

4.4 Acceptance of preproduction tests - The result of each test shall be compared with the specification requirements In the event of failure to conform to this specification, the contractor shall correct the cause of failure in the preproduction sample and on all production units The preproduction unit shall satisfactorily qualify in all tests in Groups A, B and C and shall be conducted at a laboratory satisfactory to the bureau or agency concerned. The tests shall be performed in general in the order listed.

Test	Requirement Paragraph	Test Paragraph
Group A tests		
General examination	3.3	MIL-I-983
Operating	3.3	MIL-I-983
Dielectric Strength	3.3	MIL-I-983
Insulation resistance	3.3	MIL-I-983
Group B tests		
Suppline line voltage and frequency variation	3.3	MIL-I-983
Group C tests		
Structure borne noise (vibration) for submarine equipment	3.4.6	MIL-E-22843
Airborne audible noise	3.4.6	MIL-E-22843
Shielding and radio frequency noise	3.3	MIL-I-983
Temperature and humidity	3.3	MIL-I-983
Inclination	3.3	MIL-I-983
Accelerated life	3.3	MIL-I-983
Enclosure	3.3	MIL-I-983
Vibration	3.3	MIL-I-983
Shock	3.3	MIL-I-983

4.4.1 Acceptance inspection tests - All production equipment shall be subjected to the tests in Group A at the place of manufacture

4.5 Test procedures -

4.5.1 General examination - The general examination shall be conducted in accordance with MIL-I-983.

4.5.2 Operating test - The operating test shall consist of electrically connecting the input of the speed converter to a suitable synchro as a dummy director and the output to a suitable synchro as a dummy load and energizing all circuits with rated power The output synchros should read zero when the input synchro is set on zero The input synchros should be rotated over the entire range, 0 to 40 knots to check direction of rotation and accuracy of all output signals Upon completion, the equipment shall be checked for conformance with 3.4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:54Z
Check the source to verify that this is the current version before use.

4.5.3 Accelerated life test. - The accelerated life test shall be as specified in MIL-I-983. The equipment shall be operated over the range of 0 - 40 knots oscillating from 0 to full scale and back at the rate of one or more cycles per minute.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of shipboard repair parts.

5.1.1 Basic equipment -

5.1.1.1 Preservation and packaging - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, contract or order number, contractor's name and destination.

5.1.2 Shipboard repair parts - The shipboard repair parts shall be preserved and packaged level A, packed level C and marked levels A and C respectively in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment requirements (See 6.2).

5.2.1 Preservation, packaging, packing and marking - The equipment and accessories, repair parts and technical publications shall be preserved and packaged levels A or C, packed levels A or B as specified (see 6.2) and marked in accordance with MIL-E-17555.

6. NOTES

6.1 Ordering data - Procurement documents should specify the following:

(a) Title, number, and data of this specification.
(b) Type of equipment require (see 1.2).
(c) Quantity of items required.
(d) Quantity of technical manuals required.
(e) Supply demand control point for stock repair parts.
(f) Power supply to the individual equipment (see MIL-I-983).
(g) Material and degree of enclosure (see MIL-I-983)
(h) Whether unit stock number is to be entered on the identification-plate (see MIL-I-983)
(i) Number and type of preproduction samples required for inspection.
(j) Laboratory where preproduction inspection shall be conducted
(k) Level of packaging, packing, and marking required (see 5.1)

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation the United States Government thereby incurs no responsibility or any obligation whatsoever, and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacturer use, or sell any patented invention that may in any way be related thereto.

Preparing activity
Navy - Ships
(Project 6605-N052Sh)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:54Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

INSTRUCTIONS

BUDGET BU. NO. 45-R309

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch Washington 25, D C

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a GIVE PARAGRAPH NUMBER AND WORDING

b RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES" IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU

SUBMITTED BY *(Print name and activity)*	DATE

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:54Z
Check the source to verify that this is the current version before use.

Fold

DEPARTMENT OF THE NAVY
BUREAU OF SHIPS
WASHINGTON 25 D. C.
OFFICIAL BUSINESS

POSTAGE AND FEES PAID
NAVY DEPARTMENT

CHIEF, BUREAU OF SHIPS
SPECIFICATIONS AND STANDARDIZATION BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON 25, D C

Fold

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:54Z
Check the source to verify that this is the current version before use.